DEVELOPER REFERENCE MANUAL

Carbon Fields Implementation & Best Practices

Complete Architectural Manual: Creation, Modification, Frontend Fetching & Multisite Rules

Project: Ministry of Home Affairs — Sri Lanka

Active Theme: `homeaffairs-theme`

Engine: Carbon Fields 3.x (with Awesome Extensions)

Architecture: WordPress Multisite (`/en/`, `/si/`, `/ta/`)

TABLE OF CONTENTS

1. Carbon Fields Architecture & Foundation
   Bootstrapping, hooks, containers & field packages
2. Creating Custom Fields Manually (Examples)
   Text, repeaters, media, number & Font Awesome icons
3. Modifying Existing Custom Fields (Examples)
   Adding fields, non-destructive renaming & DB structure
4. Fetching Data on the Frontend Manually
   Post meta, theme options, escaping & repeaters loop
5. Essential Rules & Best Practices
   No-seeder policy, multisite inheritance & Font Awesome
6. Quick Reference Cheat Sheet
   API functions, field types & code templates

SECTION 1

Carbon Fields Architecture & Foundation

In the `homeaffairs-theme`, custom fields are strictly decoupled from fragile UI page builders. Content structures are built natively with **Carbon Fields 3.x** and loaded via Composer with official extension packages (`htmlburger/carbon-field-icon` and `htmlburger/carbon-field-number`).

1.1 Bootstrapping Lifecycle

Carbon Fields is booted on `after_setup_theme` in `functions.php`, and all custom containers are registered through the standard hook: `carbon_fields_register_fields`.

`functions.php / inc/custom-fields.php`

```
use Carbon_Fields\Container;
use Carbon_Fields\Field;

// 1. Bootstrapping in functions.php
add_action( 'after_setup_theme', 'crb_initialize_carbon_fields' );
function crb_initialize_carbon_fields() {
    \Carbon_Fields\Carbon_Fields::boot();
}

// 2. Registering fields hook
add_action( 'carbon_fields_register_fields', 'ha_register_carbon_fields' );
function ha_register_carbon_fields() {
    // Containers defined here
}
```

1.2 Supported Container Types in the Theme

Container Type	Hook & Target Usage	Theme Example File
`post_meta`	Attached to Custom Post Types or specific Page Templates	`inc/custom-fields.php`
`theme_options`	Global settings panel under WordPress *Appearance > Theme Options*	`inc/theme-options.php`
`term_meta`	Taxonomy term metadata (e.g. category colors or banners)	`inc/custom-fields.php`

Container Type	Hook & Target Usage	Theme Example File
`nav_menu_item`	Custom attributes attached directly to WP Menu items	`inc/menu-icons.php`

SECTION 2

Creating Custom Fields Manually (With Examples)

Every field must be added inside a container. The container specifies *where* the fields appear in the WordPress administration dashboard.

2.1 Basic Container on a Specific Page Template

To attach fields to a specific page template (e.g. `page-templates/template-home.php`):

`inc/custom-fields.php`

```
Container::make( 'post_meta', __( 'News & Announcements Section', 'homeaffairs' ) )
    ->where( 'post_type', '=', 'page' )
    ->where( 'post_template', '=', 'page-templates/template-home.php' )
    ->add_fields( array(
        Field::make( 'text', 'ha_home_news_title', __( 'Section Title', 'homeaffairs'
) )
            ->set_help_text( __( 'Main section heading.', 'homeaffairs' ) ),

        Field::make( 'text', 'ha_home_news_subtitle', __( 'Section Sub Title', 'homea
ffairs' ) )
            ->set_help_text( __( 'Small pill badge above the section title.', 'homeaf
fairs' ) ),

        Field::make( 'text', 'ha_home_news_view_all_text', __( 'View All Button (Tex
t)', 'homeaffairs' ) )
            ->set_attribute( 'placeholder', __( 'View All News', 'homeaffairs' ) ),

        Field::make( 'text', 'ha_home_news_view_all_url', __( 'View All News Button L
ink (Link to Page)', 'homeaffairs' ) )
            ->set_help_text( __( 'Destination URL for the "View All News" button.',
'homeaffairs' ) ),
    ) );
```

2.2 Specialized Field Types & Usage Patterns

A. Font Awesome Icon Field (Strictly Font Awesome)

Uses the `htmlburger/carbon-field-icon` package. Always call `->add_fontawesome_options()` to ensure icons use Font Awesome:

```
Field::make( 'icon', 'qa_icon', __( 'Icon (Font Awesome)', 'homeaffairs' ) )
    ->add_fontawesome_options()
```

```
    ->set_help_text( __( 'Select Font Awesome icon from the visual icon picker.', 'ho
meaffairs' ) )
```

B. Numeric Field with Constraints

Uses the `htmlburger/carbon-field-number` package:

```
Field::make( 'number', 'ha_home_notices_count', __( 'Number of Notices to Show', 'hom
eaffairs' ) )
    ->set_min( 1 )
    ->set_max( 10 )
    ->set_step( 1 )
    ->set_attribute( 'placeholder', '3' )
    ->set_help_text( __( 'How many notices to display in the hero column (1-10).', 'h
omeaffairs' ) )
```

C. Media Gallery & File Attachments

```
// Single Image attachment (stores attachment ID)
Field::make( 'image', 'news_featured_image', __( 'Featured Image', 'homeaffairs' ) )
    ->set_help_text( __( 'Upload 1600x980 photograph.', 'homeaffairs' ) ),

// PDF / File attachment (stores attachment ID)
Field::make( 'file', 'news_attachment', __( 'Upload PDF File', 'homeaffairs' ) )
    ->set_type( array( 'pdf', 'doc', 'docx' ) )
    ->set_help_text( __( 'Direct PDF download file.', 'homeaffairs' ) )
```

2.3 Complex Repeaters (Multi-row Data)

Complex fields allow users to build repeatable rows of data, such as the homepage hero slider or statistical highlight counters.

inc/custom-fields.php (Stats Bar Repeater Example)

```
Field::make( 'complex', 'ha_home_stats', __( 'Stats Bar Items', 'homeaffairs' ) )
    ->set_layout( 'tabbed-horizontal' )
    ->add_fields( array(
        Field::make( 'text', 'stat_title', __( 'Title', 'homeaffairs' ) )
            ->set_help_text( __( 'Metric title (e.g. District Secretariats).', 'homea
ffairs' ) ),

        Field::make( 'text', 'stat_count', __( 'Count', 'homeaffairs' ) )
            ->set_help_text( __( 'Counter value (e.g. 25, 331, 14,022).', 'homeaffair
s' ) ),
    ) )
```

```
    ->set_header_template(
        '<%- stat_count ? stat_count + " - " : "" %><%- stat_title ? stat_title : "St
at " + ($_index + 1) %>'
    )
```

SECTION 3

Modifying Existing Custom Fields Manually (With Examples)

Modifying custom fields in production requires understanding how Carbon Fields stores data in the WordPress database (`wp_postmeta` and `wp_options`).

> **Critical Database Rule:** Carbon Fields prefixes all meta keys with a single underscore (_) in the database. A field defined as `ha_home_notices_btn_text` is stored as `_ha_home_notices_btn_text` in `wp_postmeta`.

3.1 Adding a New Field to an Existing Container

When adding a new field to an established container, append the field without modifying existing keys:

1. **Locate the container:** Open `inc/custom-fields.php` and find the target container (e.g. `ha_container_special_notices`).
2. **Declare the new field:** Add the new field into the `add_fields( array( ... ) )` definition:

```
// Adding Column Button (Text) to Special Notices Column
Field::make( 'text', 'ha_home_notices_btn_text', __( 'Column Button (Text)', 'homea
ffairs' ) )
    ->set_attribute( 'placeholder', __( 'View all notices', 'homeaffairs' ) )
    ->set_help_text( __( 'Button text label for the link at the bottom of the notic
es column.', 'homeaffairs' ) ),
```

3. **Update the frontend template:** Read the new key in `page-templates/template-home.php` with a graceful fallback to preserve existing layouts until saved.

3.2 Renaming a Field with Zero Data Loss (Aliasing)

If a field key must be renamed (e.g. from `qa_button_text` to `qa_btn_text`), never simply change the key in PHP, or all existing values will become inaccessible in the admin and frontend.

The Safe 2-Tier Aliasing Pattern:

```
// Step 1: Read new key first, fallback to old key if empty
$button_text = carbon_get_post_meta( $post_id, 'qa_btn_text' );
if ( empty( $button_text ) ) {
    $button_text = carbon_get_post_meta( $post_id, 'qa_button_text' );
}

// Step 2 (Optional DB Migration): Migrate legacy keys in wp_postmeta
```

```
global $wpdb;
$wpdb->query( "
    UPDATE {$wpdb->postmeta}
    SET meta_key = '_qa_btn_text'
    WHERE meta_key = '_qa_button_text'
" );
```

3.3 Complex Repeaters Storage Format

Carbon Fields stores complex repeater items using indexed delimited keys. Understanding this layout is essential for manual database queries or migrations:

Meta Key Structure in `wp_postmeta`	Description & Stored Value
`_ha_home_slides|||0|value`	Row index marker (value: `_`)
`_ha_home_slides|slide_title|0|0|value`	Field `slide_title` of Row 0
`_ha_home_slides|slide_btn_text|0|0|value`	Field `slide_btn_text` of Row 0
`_ha_home_slides|slide_title|1|0|value`	Field `slide_title` of Row 1

SECTION 4

Fetching Data to Front End Using Carbon Fields Manually

4.1 Core Retrieval Functions

Function	Usage Scope	Syntax Example
`carbon_get_post_meta( $id, $name )`	Posts, Pages, CPTs	`$title = carbon_get_post_meta( get_the_ID(), 'news_custom_title' );`
`carbon_get_theme_option( $name )`	Global Theme Options	`$phone = carbon_get_theme_option( 'ha_telephone' );`
`carbon_get_term_meta( $id, $name )`	Taxonomy Terms	`$color = carbon_get_term_meta( $term_id, 'cat_color' );`

4.2 Practical Frontend Examples with Escaping

Example 1: Text, Link & Button with Conditionals

```
<?php
$page_id            = get_the_ID();
$news_sec_title     = carbon_get_post_meta( $page_id, 'ha_home_news_title' );
$news_view_all_text = carbon_get_post_meta( $page_id, 'ha_home_news_view_all_text' );
$news_view_all_url  = carbon_get_post_meta( $page_id, 'ha_home_news_view_all_url' );

// Apply localized fallback if empty
$display_btn_text = ! empty( $news_view_all_text ) ? $news_view_all_text : __( 'View
All News', 'homeaffairs' );
?>

<?php if ( ! empty( $news_sec_title ) ) : ?>
  <h2 class="section-title"><?php echo esc_html( $news_sec_title ); ?></h2>
<?php endif; ?>

<?php if ( ! empty( $news_view_all_url ) ) : ?>
  <a href="<?php echo esc_url( $news_view_all_url ); ?>" class="btn btn-primary">
    <?php echo esc_html( $display_btn_text ); ?>
    <i class="fas fa-arrow-right" aria-hidden="true"></i>
  </a>
<?php endif; ?>
```

Example 2: Resolving Image & File URLs

Image and file fields in Carbon Fields store the WordPress **Attachment ID**. Convert them to URLs using core WordPress functions:

```
<?php
$image_id = carbon_get_post_meta( $post_id, 'news_featured_image' );
$file_id  = carbon_get_post_meta( $post_id, 'news_attachment' );

// 1. Image URL with size
$image_url = $image_id ? wp_get_attachment_image_url( $image_id, 'large' ) : '';

// 2. Direct File Download URL
$file_url = $file_id ? wp_get_attachment_url( $file_id ) : '';
?>

<?php if ( $image_url ) : ?>
  <img src="<?php echo esc_url( $image_url ); ?>" alt="<?php echo esc_attr( get_the_t
itle( $post_id ) ); ?>" loading="lazy">
<?php endif; ?>

<?php if ( $file_url ) : ?>
  <a href="<?php echo esc_url( $file_url ); ?>" target="_blank" download>
    <i class="fas fa-file-pdf"></i> <?php esc_html_e( 'Download Document (PDF)', 'hom
eaffairs' ); ?>
  </a>
<?php endif; ?>
```

Example 3: Looping Through Complex Repeaters

```
<?php
$raw_slides = carbon_get_post_meta( $page_id, 'ha_home_slides' );
$slides     = ( ! empty( $raw_slides ) && is_array( $raw_slides ) ) ? $raw_slides : a
rray();
?>

<?php if ( ! empty( $slides ) ) : ?>
  <div class="hero-slider">
    <?php foreach ( $slides as $index => $slide ) :
      $slide_title = isset( $slide['slide_title'] ) ? $slide['slide_title'] : '';
      $slide_desc  = isset( $slide['slide_desc'] ) ? $slide['slide_desc'] : '';
      $slide_btn   = isset( $slide['slide_btn_text'] ) ? $slide['slide_btn_text'] :
'';
      $slide_url   = isset( $slide['slide_btn_url'] ) ? $slide['slide_btn_url'] :
'#';
      $slide_img   = isset( $slide['slide_image'] ) ? $slide['slide_image'] : '';
    ?>
      <div class="slide-item" data-index="<?php echo esc_attr( $index ); ?>">
```

```
        <?php if ( $slide_img ) : ?>
          <img src="<?php echo esc_url( $slide_img ); ?>" alt="<?php echo esc_attr(
$slide_title ); ?>">
        <?php endif; ?>
        <h3><?php echo esc_html( $slide_title ); ?></h3>
        <p><?php echo esc_html( $slide_desc ); ?></p>
        <?php if ( ! empty( $slide_btn ) ) : ?>
          <a href="<?php echo esc_url( $slide_url ); ?>" class="btn"><?php echo esc_h
tml( $slide_btn ); ?></a>
        <?php endif; ?>
      </div>
    <?php endforeach; ?>
  </div>
<?php endif; ?>
```

SECTION 5

Other Required Things to Know When Using Carbon Fields

5.1 The "No-Seeder / Keep Empty" Policy

> **Theme Policy Requirement:** When custom fields are not populated by the user, **never output fake seeder fallback data** (e.g. dummy phone numbers, test emails, or hardcoded addresses). If empty, output an empty string (`''`) or cleanly suppress the enclosing HTML block.

Why Avoid `->set_default_value()` in Carbon Fields:
In Carbon Fields, calling `->set_default_value('sample')` causes `carbon_get_theme_option()` and `carbon_get_post_meta()` to return `'sample'` even when the database option has never been saved! This breaks the no-seeder policy.

The Solution — Use Placeholders:
Instead of `->set_default_value()`, use `->set_attribute('placeholder', '...')`. This provides a helpful visual guide to content editors in the admin dashboard without injecting dummy data into queries.

```
// ❌ BAD: Returns default value even when empty in DB
Field::make( 'text', 'ha_telephone', __( 'Phone', 'homeaffairs' ) )
    ->set_default_value( '+94 112 050 450' )

// ✅ GOOD: Shows placeholder hint to admin, but returns empty string if unpopulated
Field::make( 'text', 'ha_telephone', __( 'Phone', 'homeaffairs' ) )
    ->set_attribute( 'placeholder', '+94 112 050 450' )
```

5.2 Multisite Theme Options & English Fallback

In WordPress Multisite (`/moha/`, `/moha/si/`, `/moha/ta/`, `/moha/en/`), options are saved per-site. Use the helper function `ha_get_theme_option()` in `inc/helpers.php`, which automatically inherits values from the primary English site (Blog 4) when a sub-site option is empty:

```
// Automatic English fallback in inc/helpers.php:
$phone = ha_get_theme_option( 'ha_telephone' );
// If Sinhala site has no custom phone, it inherits from English site.
// If both are empty, it returns empty string ('') with zero dummy data.
```

5.3 Font Awesome Exclusivity & Visual Icon Picker

Under theme architectural rules:

- All custom field icons **must strictly use Font Awesome** (zero Dashicons).

- Rely exclusively on Carbon Fields' built-in React visual icon picker.
- **Do NOT** inject jQuery wrapper buttons (such as custom `+ Browse Icons` modals) over Carbon Fields inputs, as this causes UI duplication and layout breakage.

5.4 Suppressing Default Core Editor & Featured Image

On page templates and CPTs where Carbon Fields manages all structured content, the default WordPress WYSIWYG editor and Featured Image metabox must be cleanly suppressed to avoid confusing editors:

`inc/custom-fields.php`

```
add_action( 'admin_init', 'ha_hide_default_editor_and_thumbnail_for_templates' );
function ha_hide_default_editor_and_thumbnail_for_templates() {
    remove_post_type_support( 'special_notice', 'editor' );
    remove_post_type_support( 'news_announcement', 'editor' );

    if ( ha_is_custom_page_template() ) {
        remove_post_type_support( 'page', 'editor' );
        remove_post_type_support( 'page', 'thumbnail' );
    }
}
```

SECTION 6

Quick Reference & Syntax Cheat Sheet

Operation	Code Snippet
Import Classes	`use Carbon_Fields\Container;` `use Carbon_Fields\Field;`
Text Field	`Field::make( 'text', 'field_key', __( 'Label', 'homeaffairs' ) )`
Textarea	`Field::make( 'textarea', 'field_key', __( 'Label', 'homeaffairs' ) )->set_rows( 4 )`
Rich Text (WYSIWYG)	`Field::make( 'rich_text', 'field_key', __( 'Label', 'homeaffairs' ) )`
Font Awesome Icon	`Field::make( 'icon', 'field_key', __( 'Label', 'homeaffairs' ) )->add_fontawesome_options()`
Number Field	`Field::make( 'number', 'field_key', __( 'Label', 'homeaffairs' ) )->set_min( 1 )->set_max( 10 )`
Image Field	`Field::make( 'image', 'field_key', __( 'Label', 'homeaffairs' ) )`
File Field (PDF)	`Field::make( 'file', 'field_key', __( 'Label', 'homeaffairs' ) )->set_type( array( 'pdf' ) )`
Conditional Logic	`->set_conditional_logic( array( array( 'field' => 'toggle', 'value' => 'yes' ) ) )`
Frontend Fetch (Post)	`$val = carbon_get_post_meta( $post_id, 'field_key' );`
Frontend Fetch (Theme Option)	`$val = ha_get_theme_option( 'field_key' );`